Exercise 1. From a standard deck of 52 cards we choose 5. In how many ways can the selection be made so that:
a) there are exactly 3 cards of the same suit
b) there is at least one face card

Exercise 2. On a plane, n straight lines are drawn so that no two lines are parallel and no three lines intersect at a single point. How many intersection points are obtained?

Exercise 3. How many ten-digit numbers are there such that the sum of their digits equals 5?

Exercise 4. A tournament involves 16 teams. In how many ways can these teams be arranged into 8 pairs?